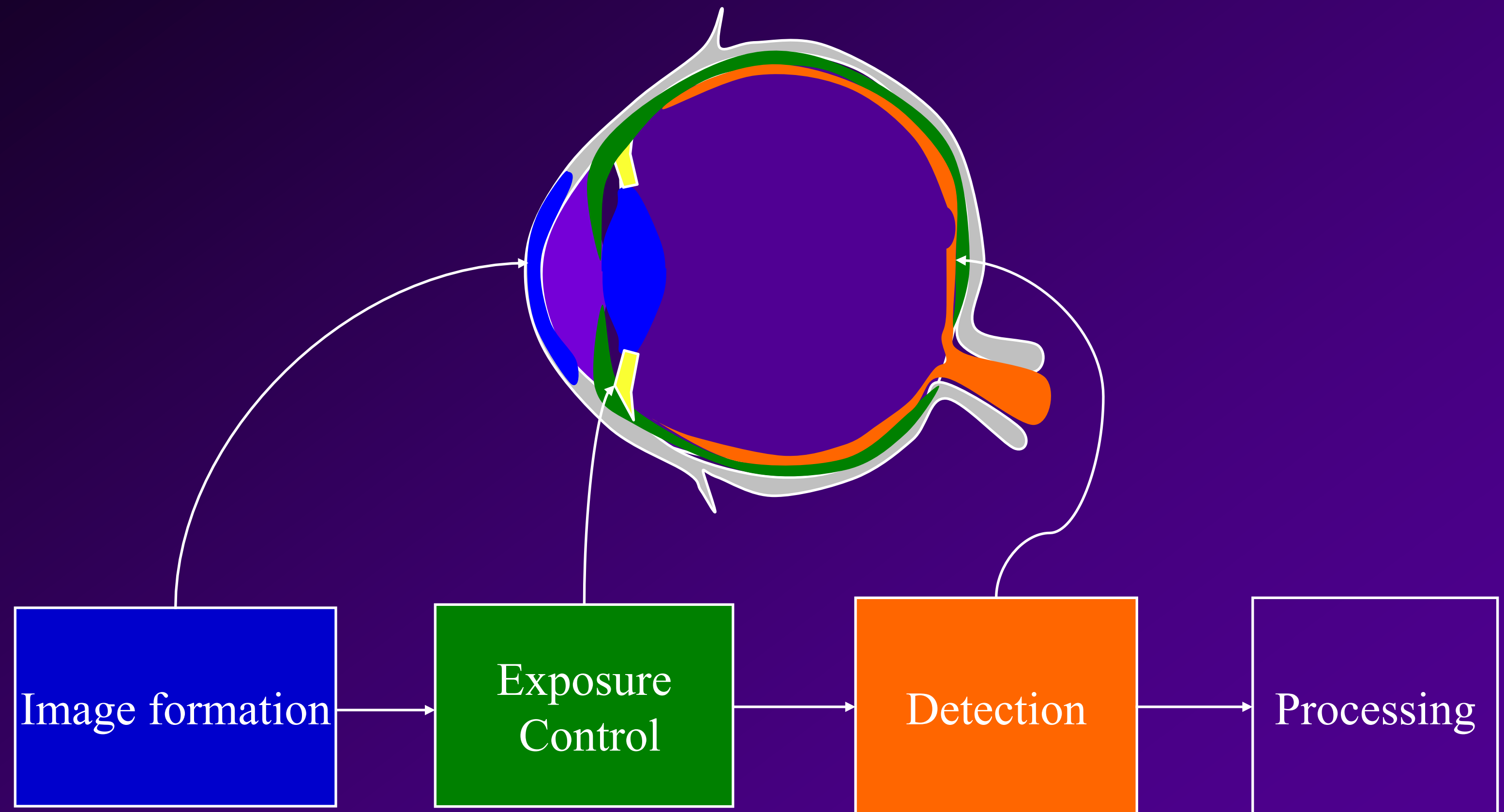
The Human Visual System

The Eye

In this section . . .

- Anatomy of human eye
- Image formation by human eye
- Method of light detection
- Retinal processing
- Eye optical defects and diseases

Human Visual System



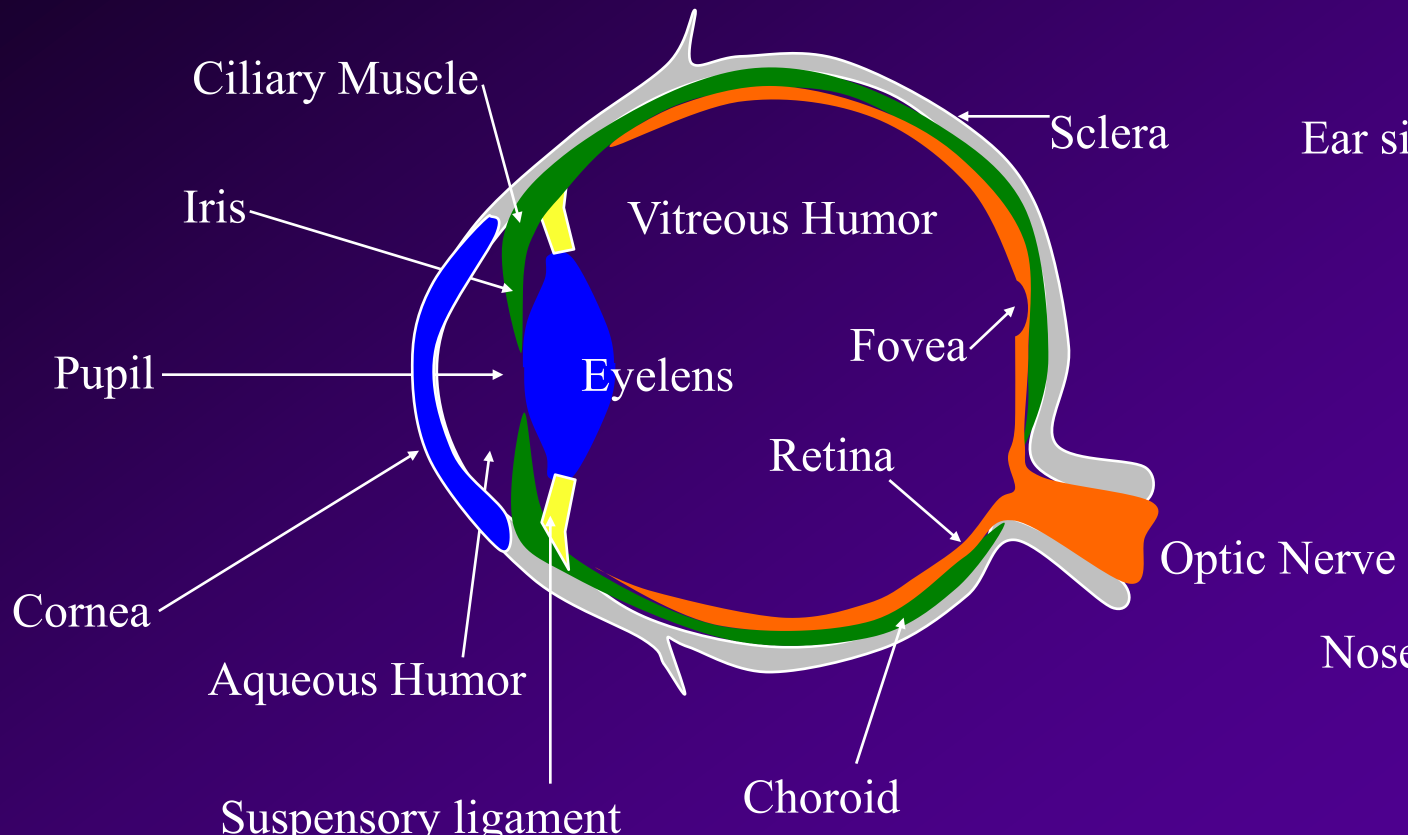
- Cornea
- Lens

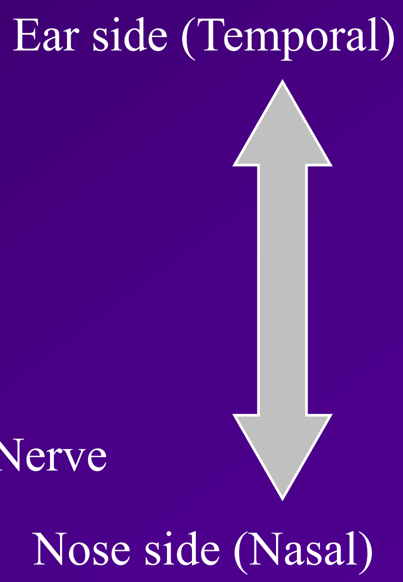
- Iris/pupil
- Photoreceptor sensitivity

- Retina
- Rods
- Cones

- Brain

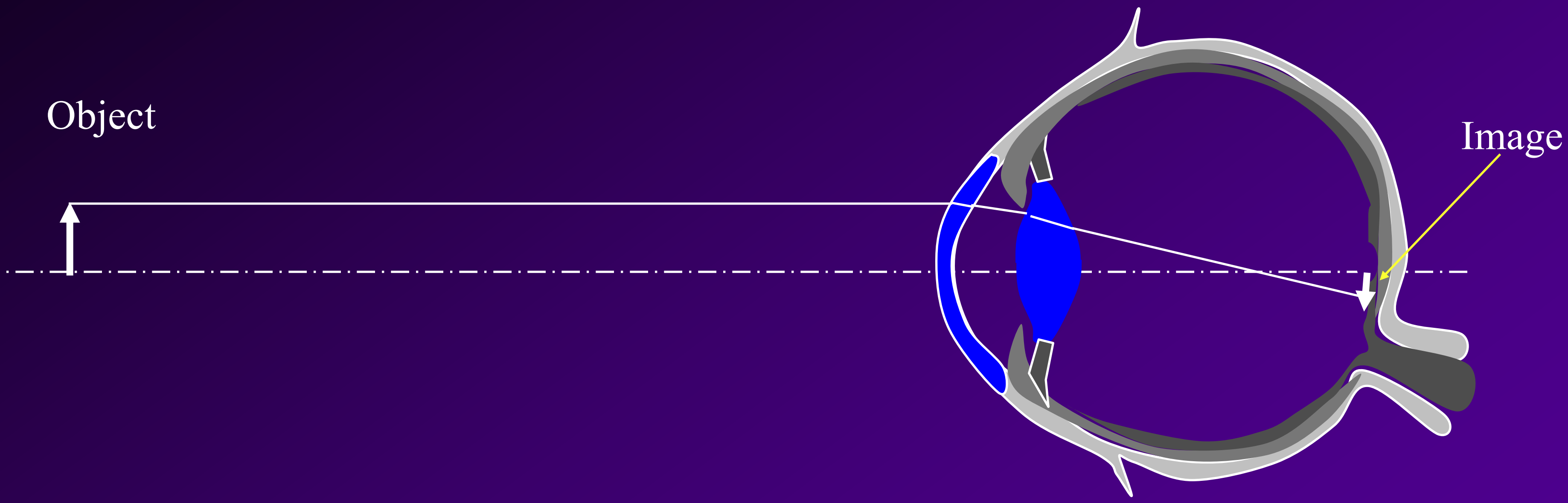
Human Eye





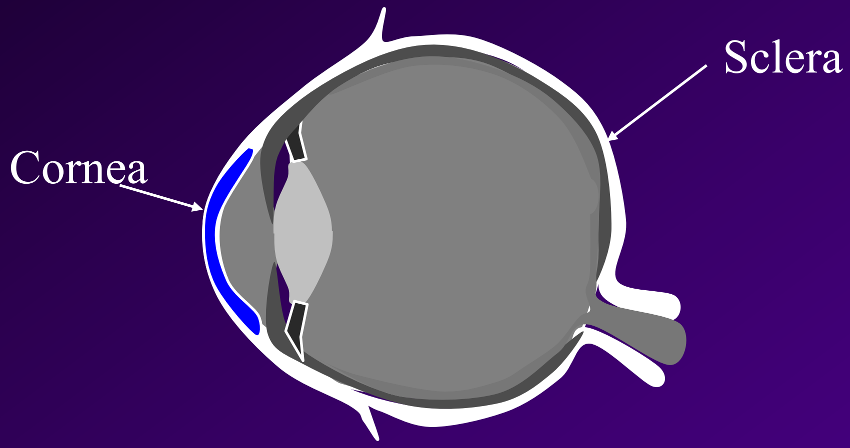
- Human eye is a complete imaging system.

Image Formation



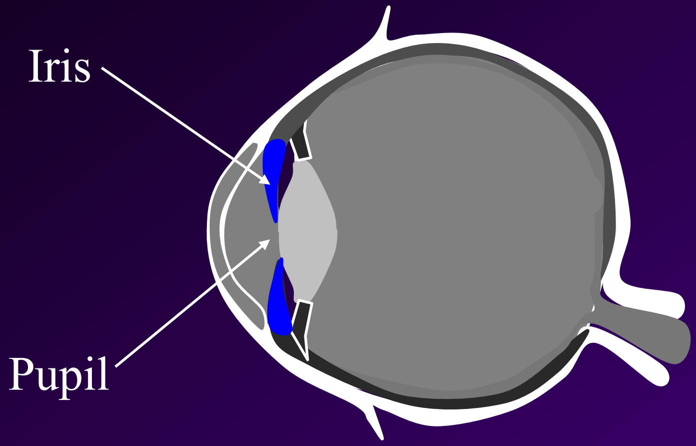
- The curved surfaces of the eye focus the image onto the back surface of the eye.

Cornea



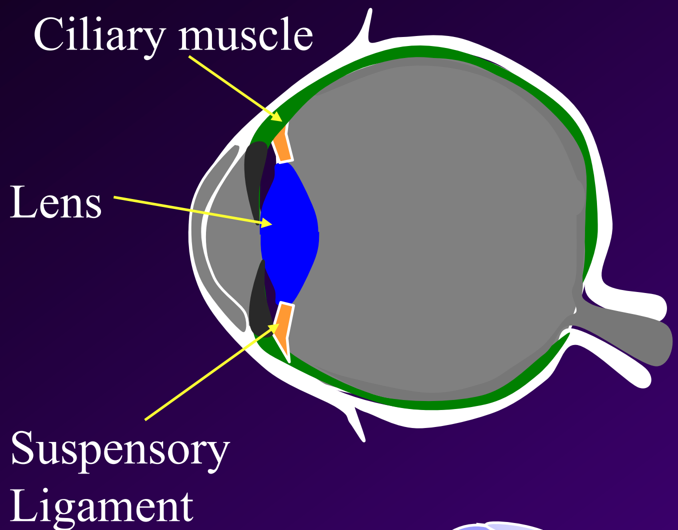
- The outer wall of the eye is formed by the hard, white *sclera*.
- *Cornea* is the clear portion of the sclera.
- 2/3 of the refraction takes place at the cornea.

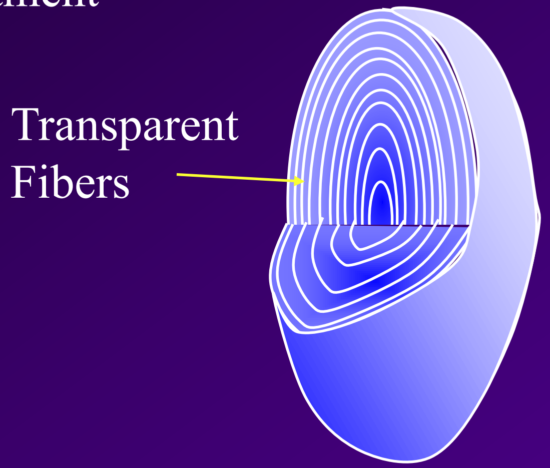
Iris and Pupil



- Colored *iris* controls the size of the opening (*pupil*) where the light enters.
- Pupil determines the amount of light, like the aperture of a camera.

Lens

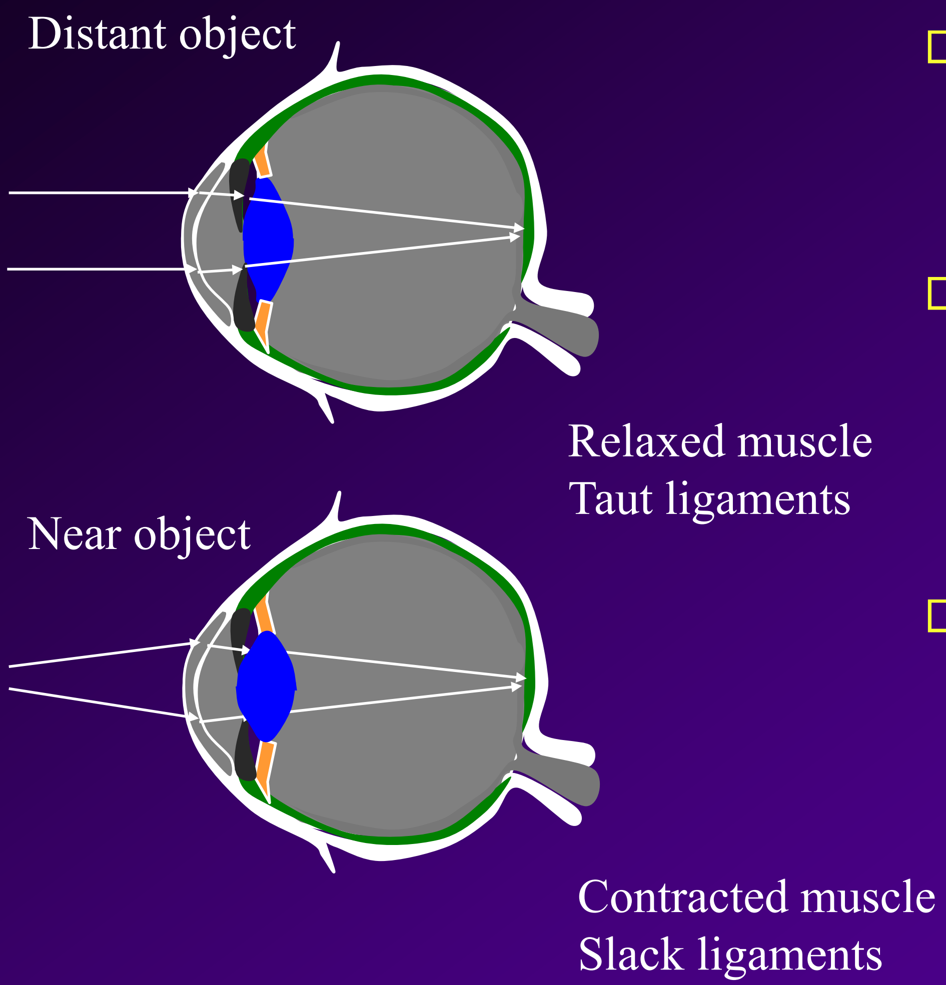




Cross section of the eye lens

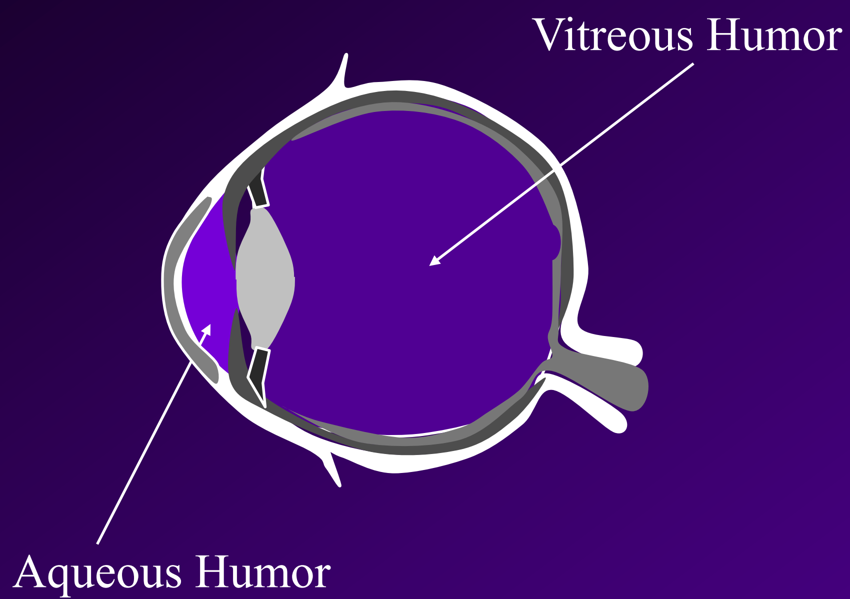
- *Eye lens* is made of transparent fibers in a clear membrane.
- Suspended by suspensory ligament.
- Used as a fine focusing mechanism by the eye; provides 1/3 of eye's total refracting power.
- Non-uniform index of refraction.

Accommodation



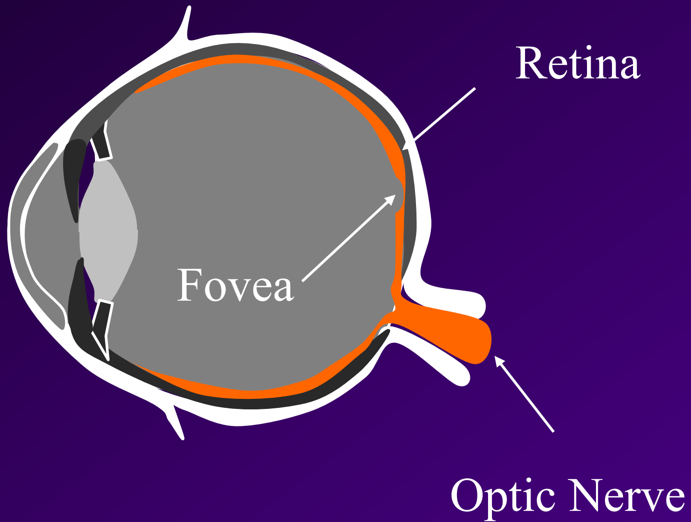
- The *suspensory ligaments* attach the lens to the *ciliary muscle*.
- When the muscle contracts, the lens bulges out in the back, decreasing its focal length.
- The process by which the lens changes shape to focus is called *accommodation*.

Aqueous Humor and Vitreous Humor



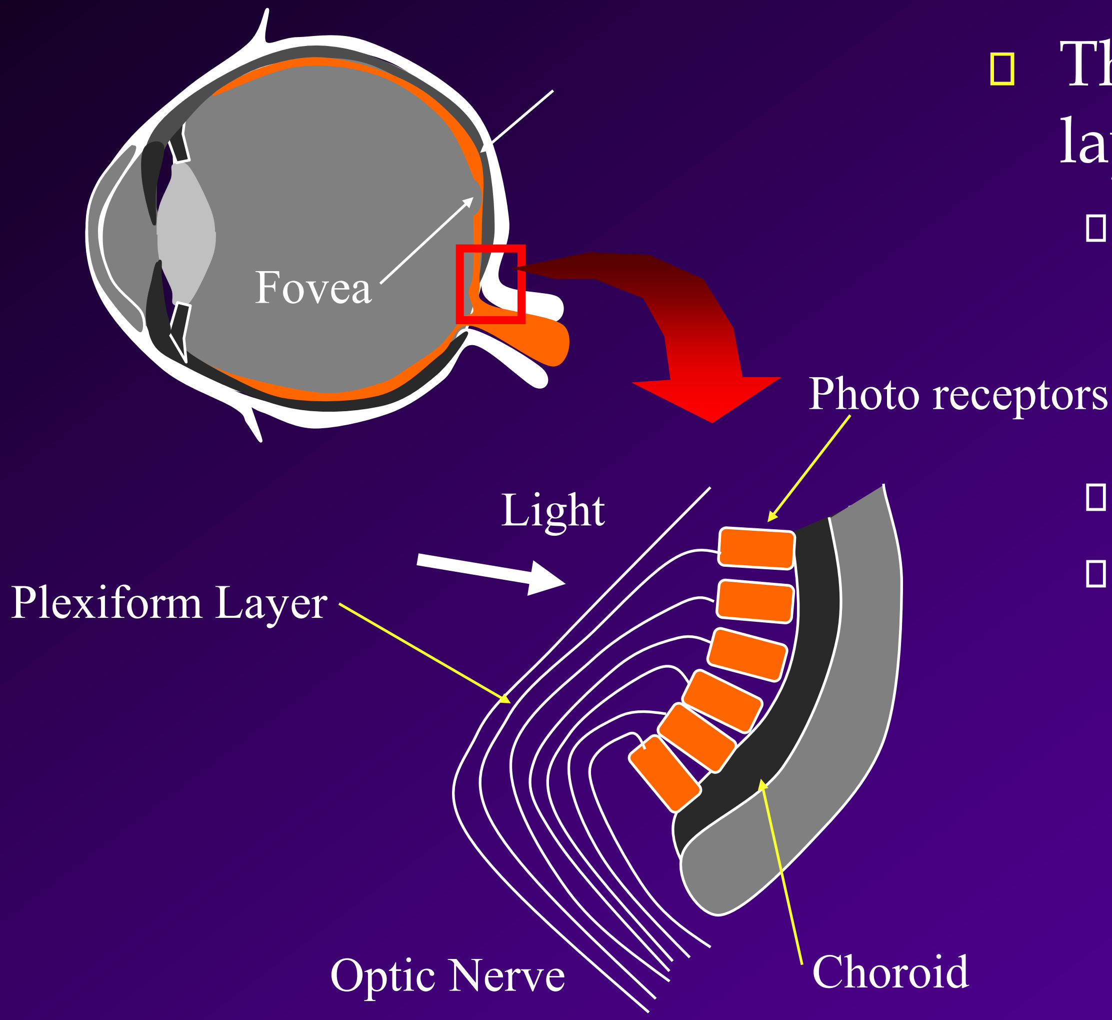
- Transparent gelatinous liquid filling the eye.
- Provides nutrients to the cornea and eye lens.
- Also helps maintain the eyeball shape with its pressure.

Retina



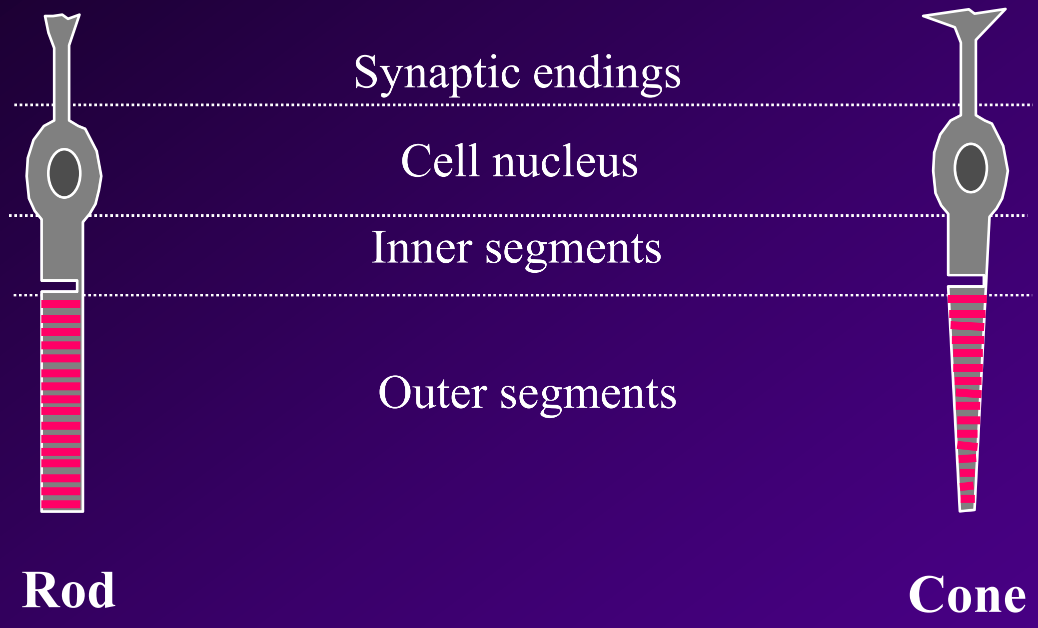
- *Retina* is the photosensitive “detector” for the eye.
- Two types of receptors in the retina: *rods* for low light level, and *cones* for color.
- Located at the center of the retina, *fovea* contains a greater concentration of cones.
- Signals from the receptors leave through the *optic nerve* to the brain.

Plexiform Layer



- The retina is made of three layers:
  - *Plexiform layer* is a network of nerves which carry the signals from the photo receptors.
  - Photo receptors.
  - *Choroid* provides nourishment to the receptors, as well as absorb any light that didn’t get absorbed by the photo receptors, like a antihalation backing in film.

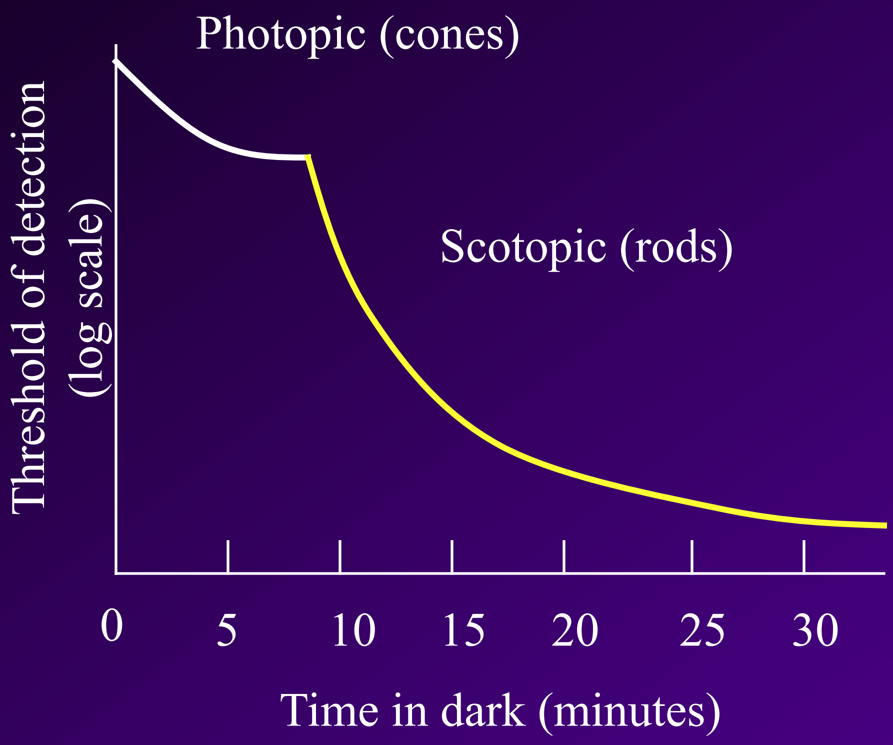
Rods and Cones



- Highly sensitive to low light level or *scotopic* conditions.
- Black and white.
- Dispersed in the periphery of the retina.

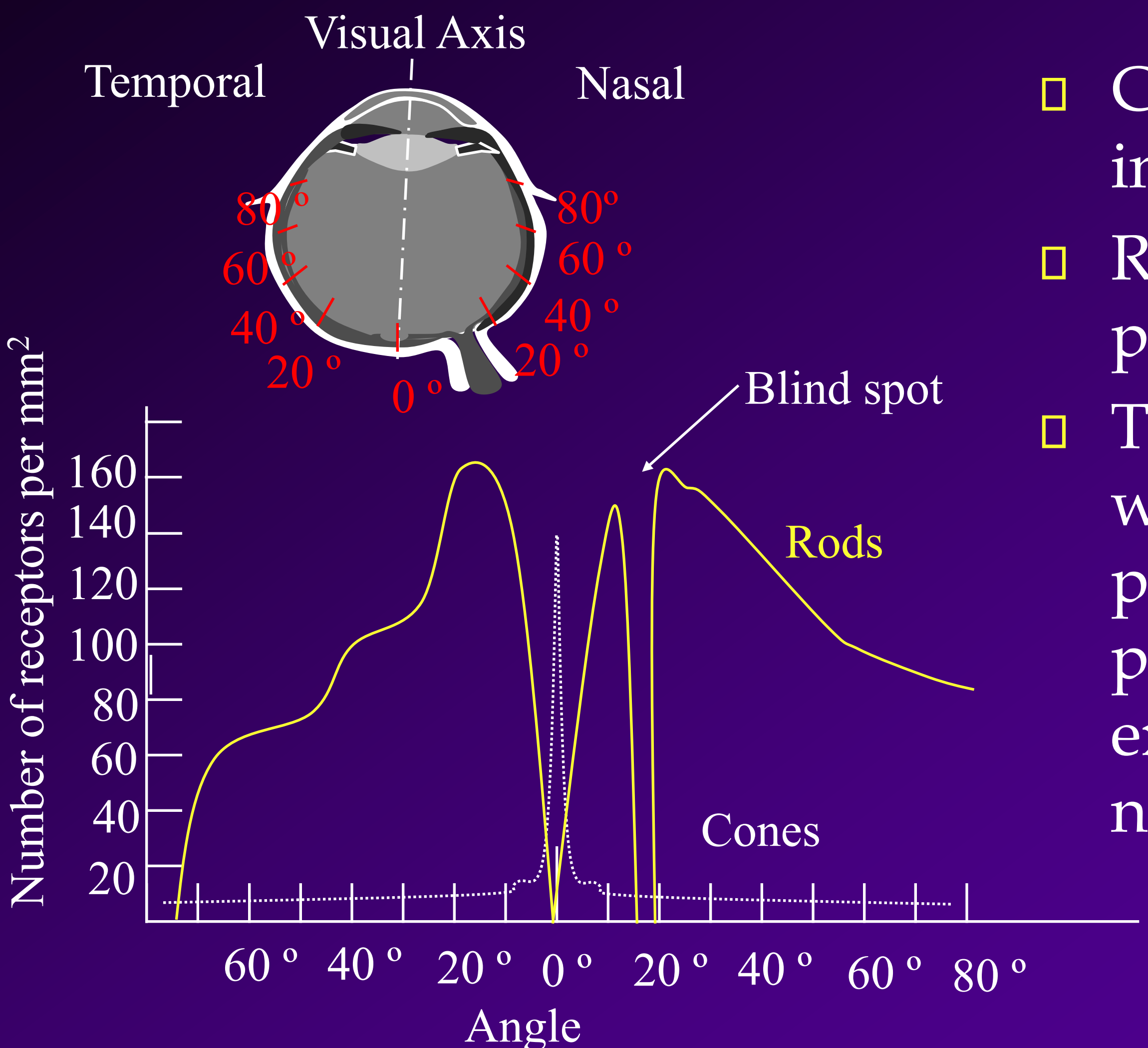
- Sensitive to high light level or *photopic* conditions.
- Three types of cones responsible for color vision.
- Concentrated in the fovea.

Adaptation



- Why can't you see immediately after you enter a movie theater from daylight?
- The threshold of detection changes with overall light level.
- The switch is quite gradual, until the sensitivities of cones and rods cross over at about 7 minutes in the dark.

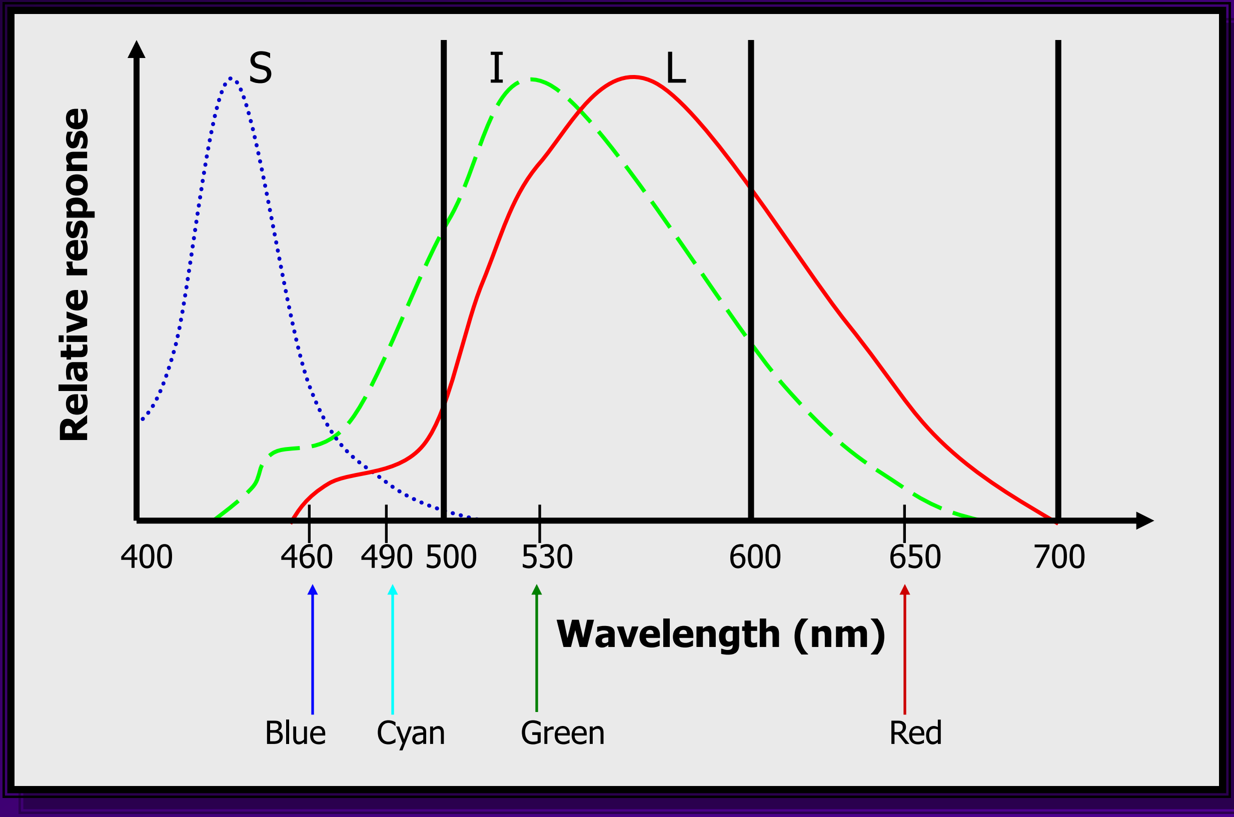
Distribution of Photoreceptors



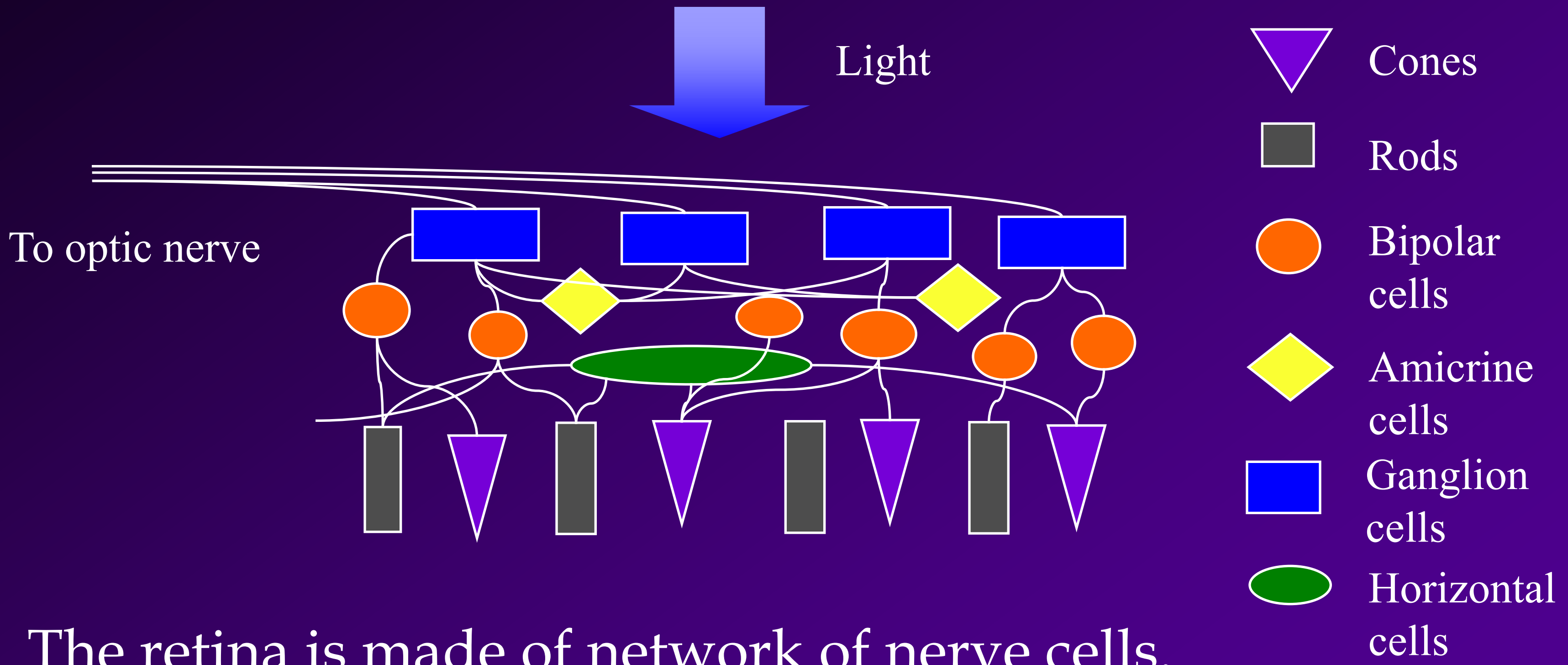
- Cones are concentrated in the fovea.
- Rods predominate the periphery.
- There is a blind spot where there are no photoreceptors, at the point where the nerves exit the eye (optic nerve).

Human Vision

- Human Cone Response to Color
  - three cone types (S,I,L) correspond to B,G,R

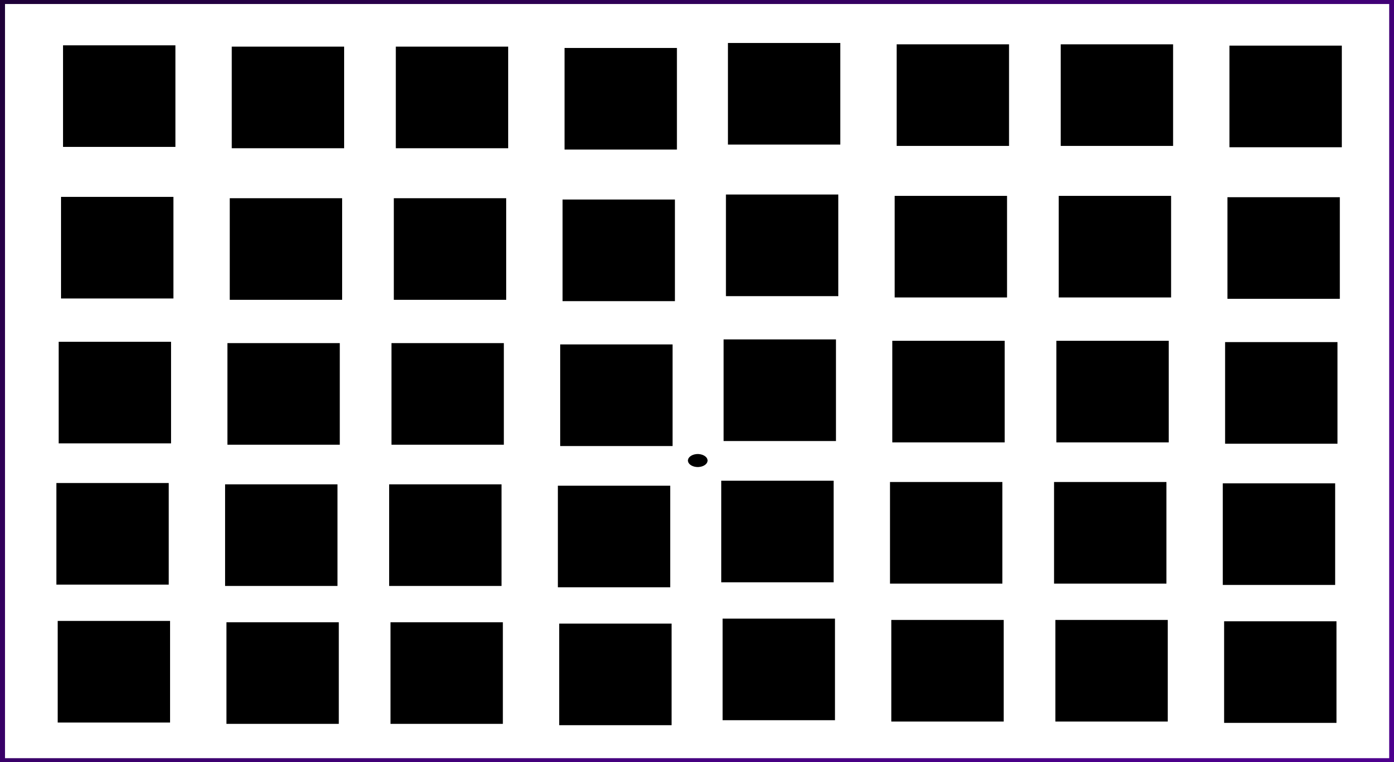


Retina



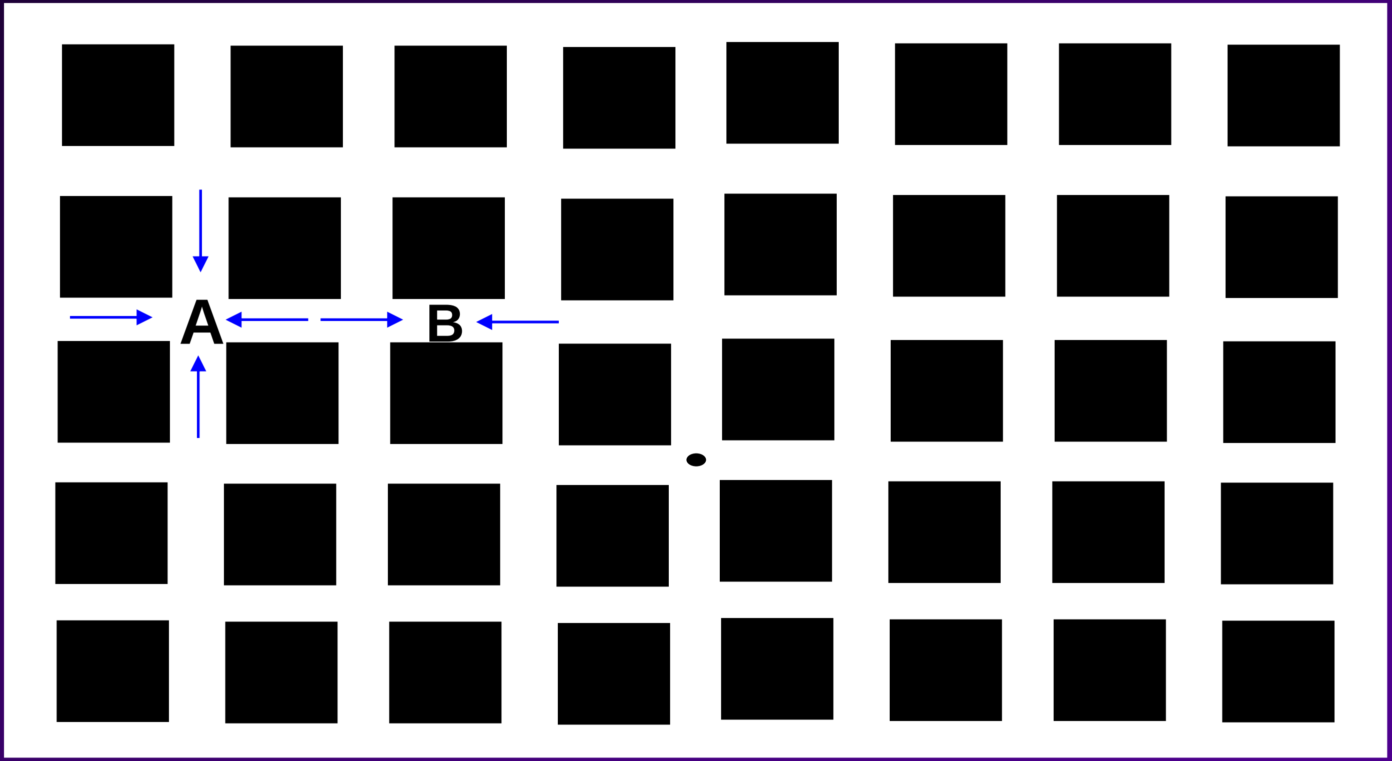
- The retina is made of network of nerve cells.
- The network works together to reduce the amount of information in a process called *lateral inhibition*.

Hermann Grid



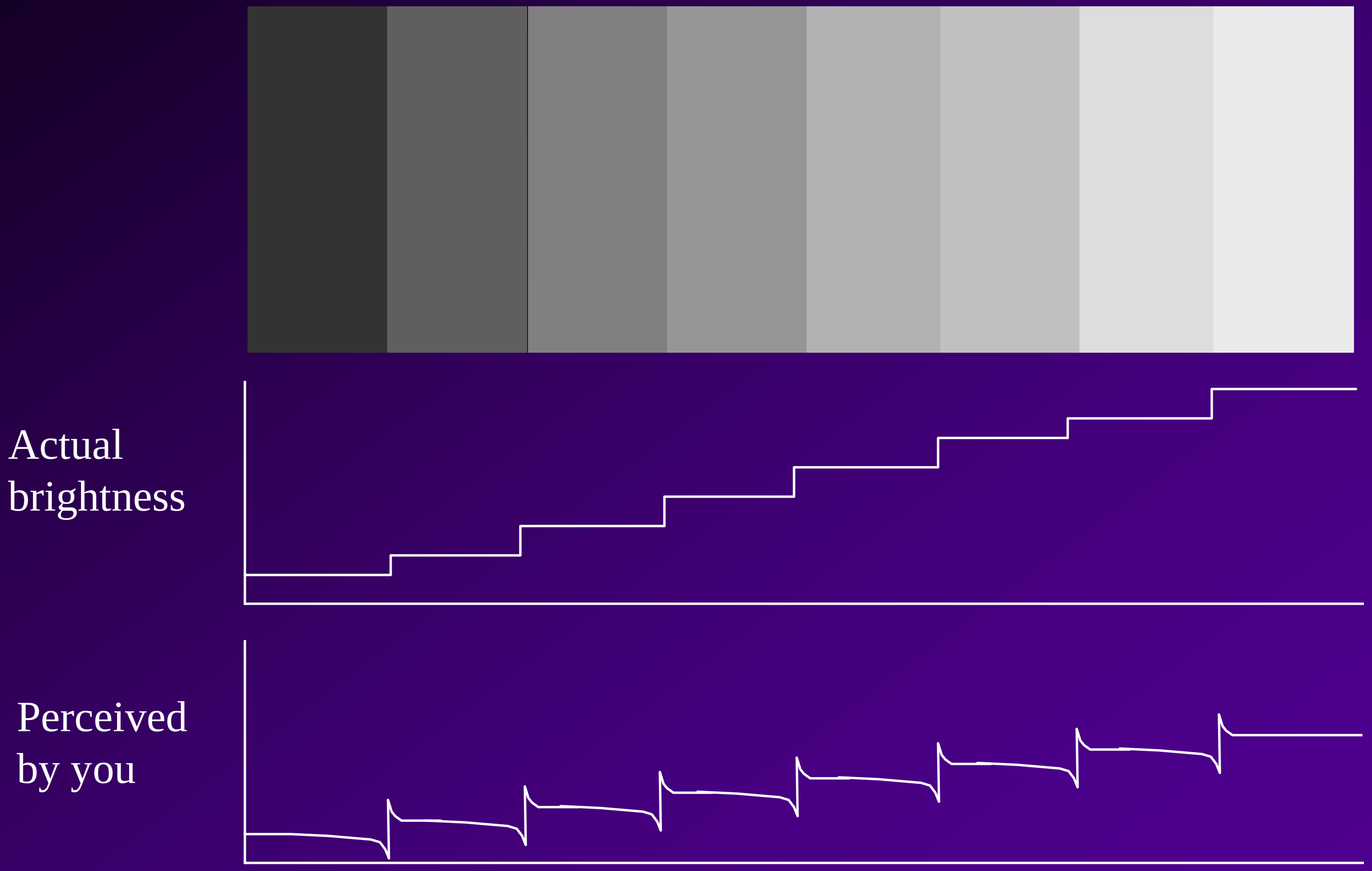
- Illustrates lateral inhibition.

Hermann Grid

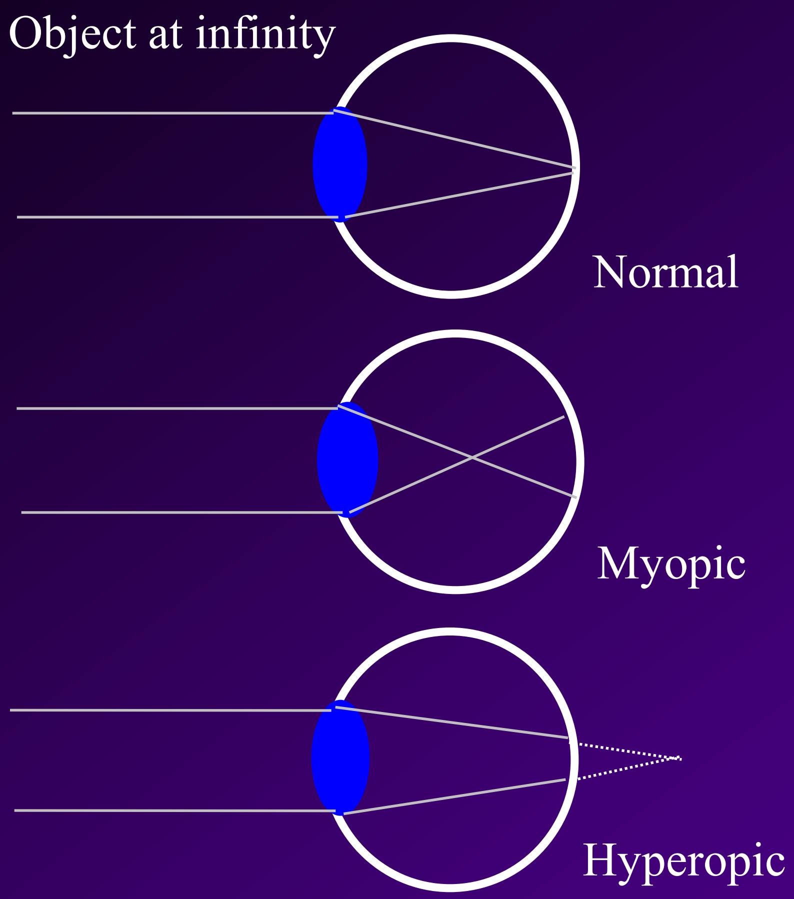


- Point A looks darker because there are 4 inhibitory inputs
- Point B looks lighter because there are only 2 inhibitory

Mach Bands



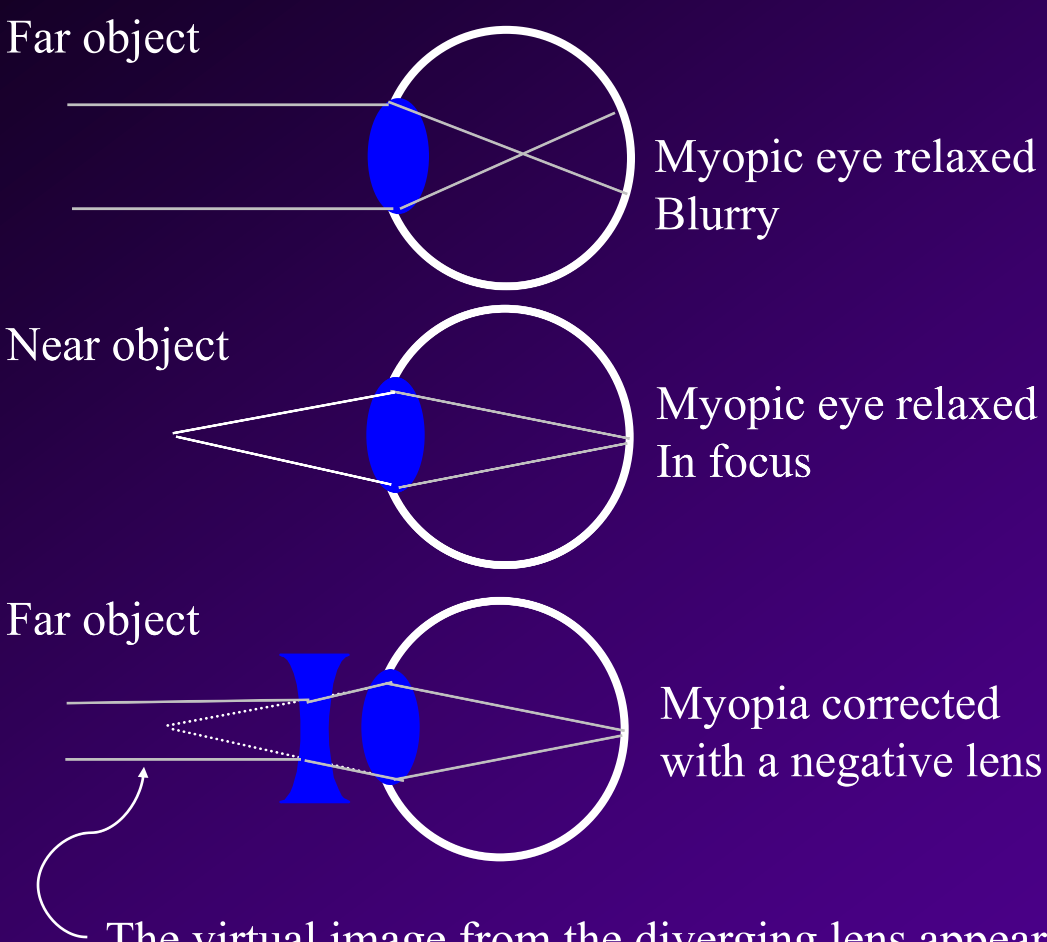
Eye Defects



Eyes at relax state.

- Image focuses on the retina for a normal eye.
- Distant objects look blurry for a *myopic* (near sighted) eye.
- Near objects look blurry for a *hyperopic* (far sighted) eye.

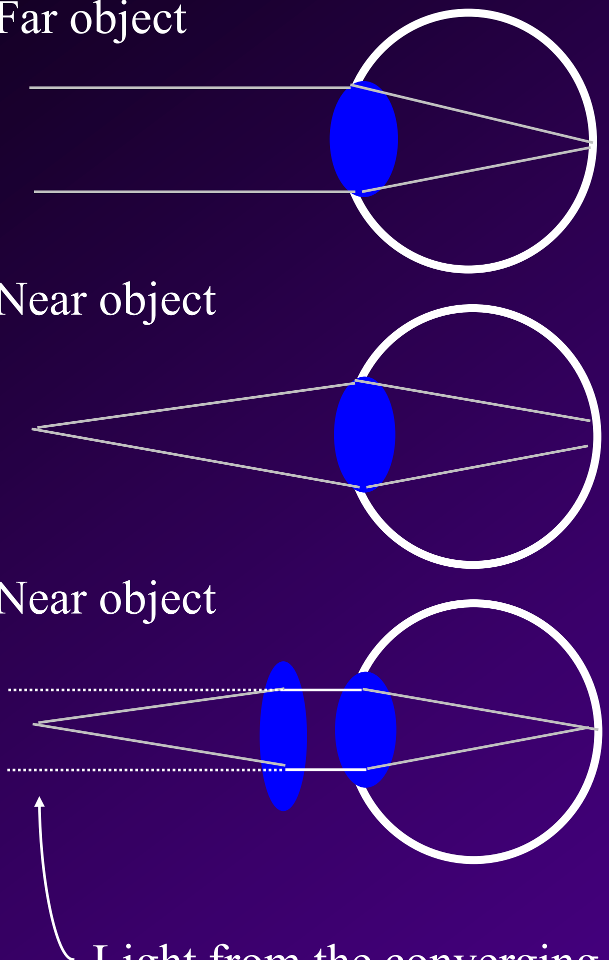
Myopia - Near sightedness



The virtual image from the diverging lens appears to be closer.

- Distant objects look blurry because the eye cannot relax any farther so that the image is focused before the retina.
- Near object in focus without accommodation.
- Corrected with a negative lens.

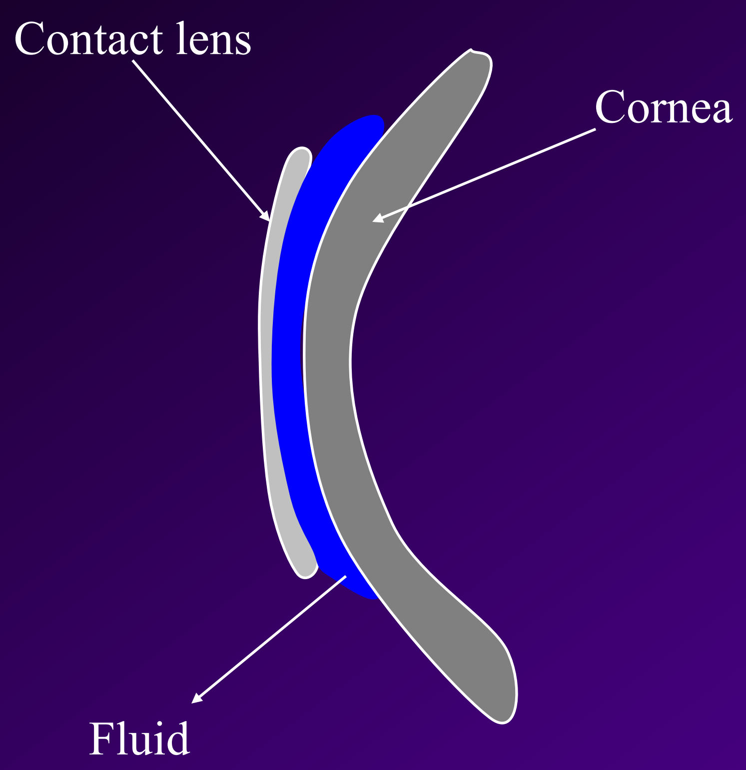
Hyperopia - Far sightedness



Light from the converging lens looks as though it is coming from the distance.

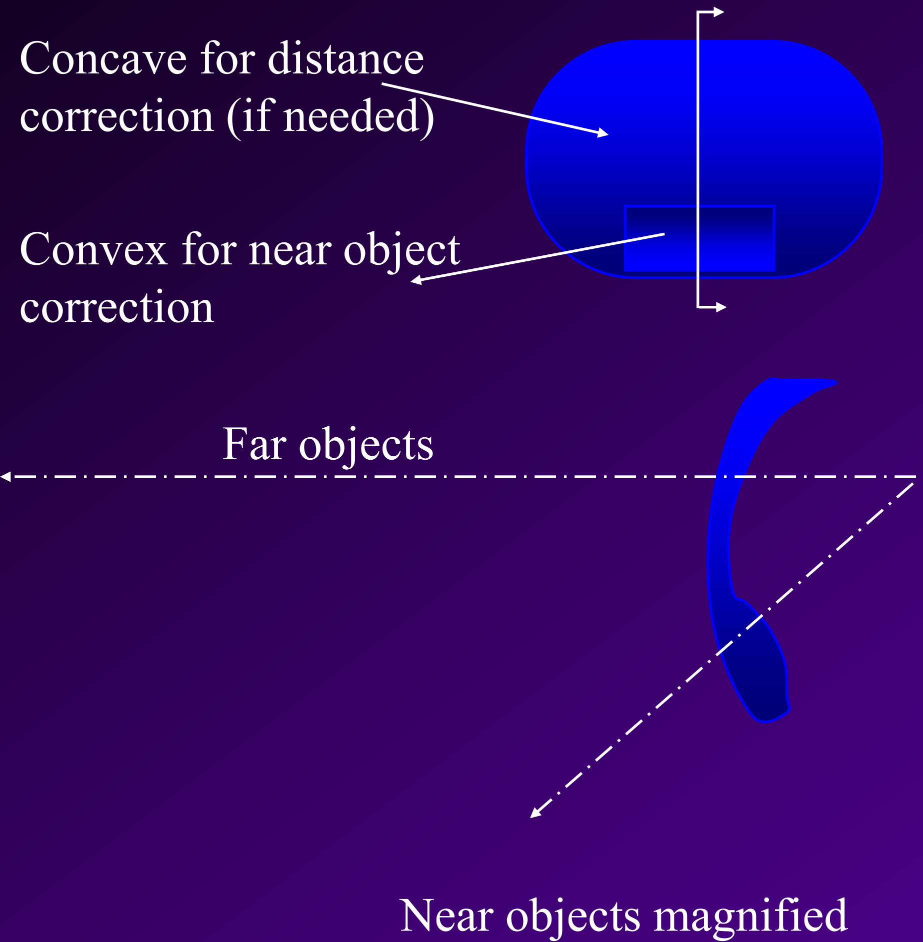
- Near objects look blurry because the eye cannot accommodate enough for near objects.
- Far object in focus.
- Corrected with a positive lens.

Contact Lens



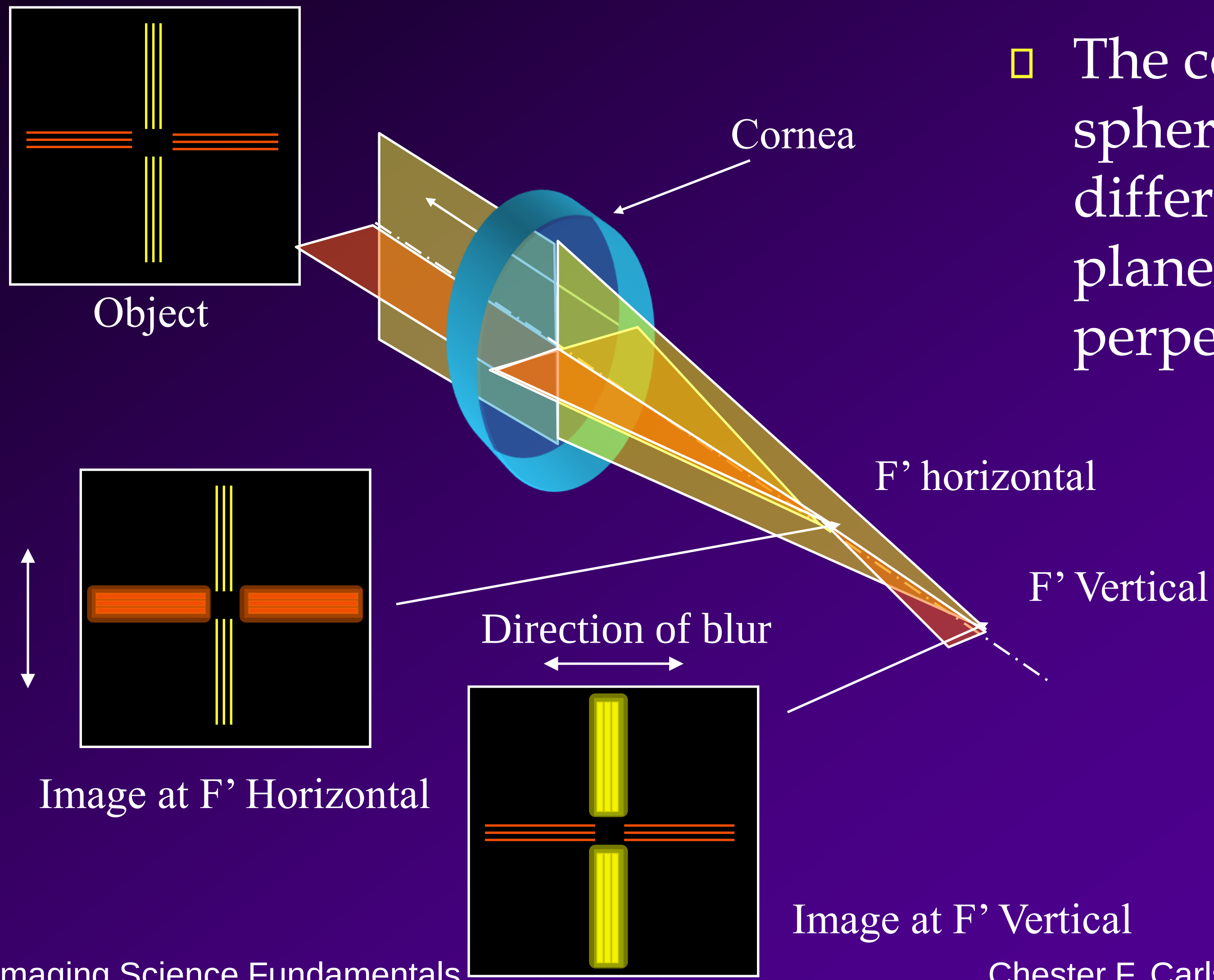
- Contact lens is an alternative to corrective lenses.
- Changes the curvature of the cornea by adhering to the surface with some fluid.

Presbyopia - "Old eye"



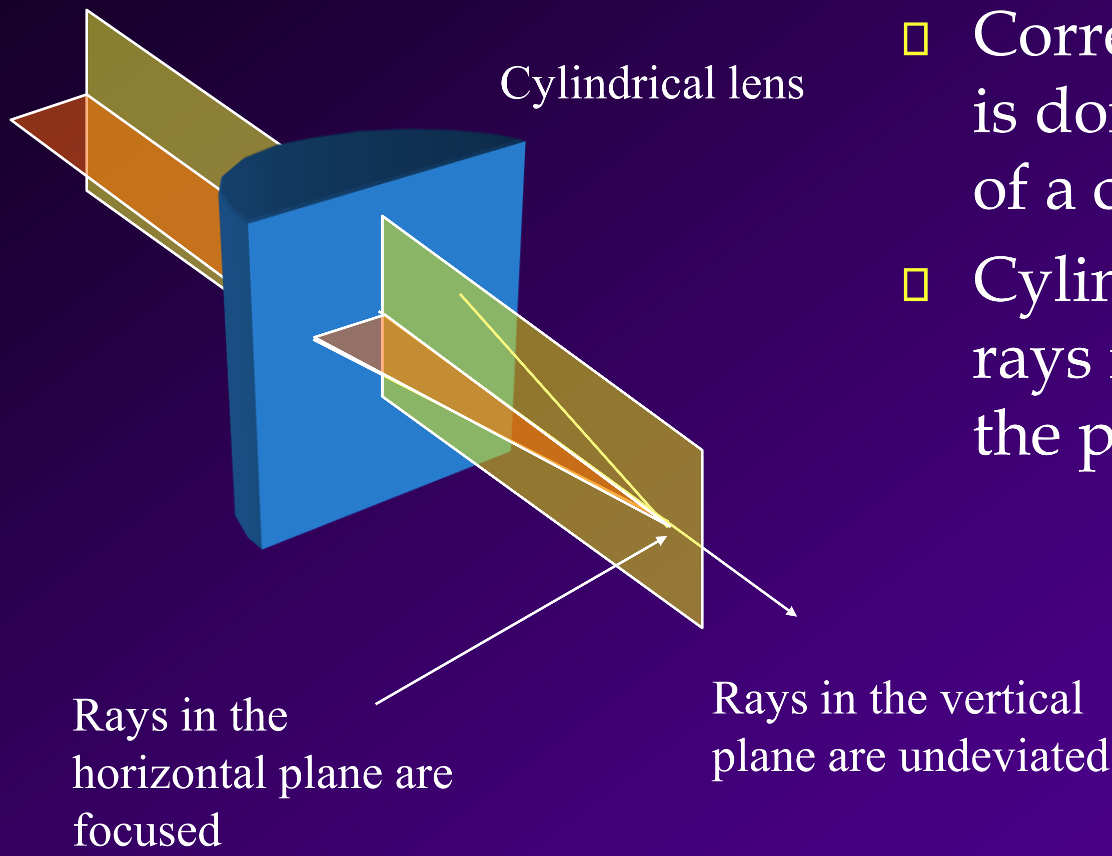
- Lens hardens with age.
- Eye cannot adequately accommodate near objects.
- Bifocals (lens with two focal lengths) contains a concave lens for distance (if needed) and a convex lens for near objects.

Astigmatism



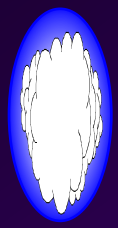
- The cornea is not spherical; Focal length different from one plane to a perpendicular plane.

Astigmatism

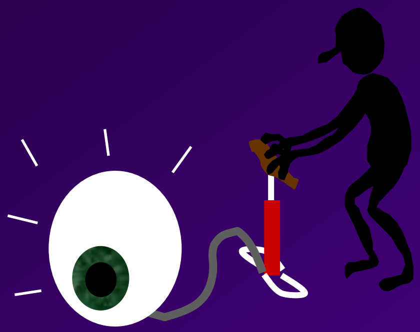


- Correction of astigmatism is done through the use of a cylindrical lens.
- Cylindrical lens converge rays in one plane but not the perpendicular plane.

Common Eye Diseases

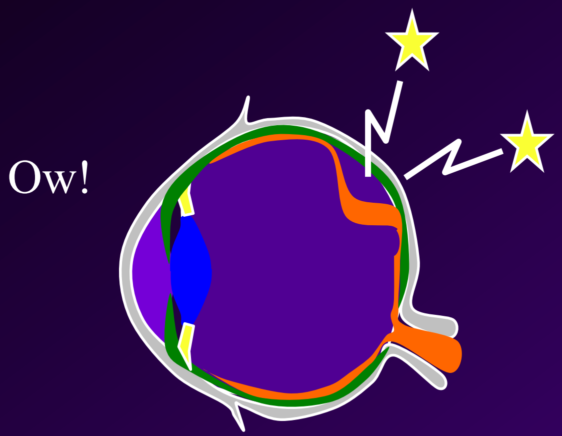


- *Cataract* - Clouding of the lens.
  - Symptom: Loss of vision
  - Cure: Lens replacement

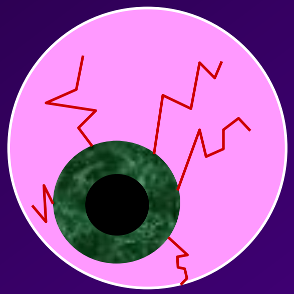


- *Glaucoma* - Pressure buildup in the eye, damaging the retina.
  - Symptom: Loss of vision first in the periphery.
  - Cure: Surgery to drain fluid from the eye.
    - Loss of vision is usually permanent

Common Eye Diseases



- *Detached retina* - portion of the retina detaches from the back of the eye.
  - Symptom: Perception of flashes, Loss of vision
  - Cure: Laser surgery to reattach retina
- *Pink eye* - Infection of the surface of the eye.
  - Symptom: Irritation
  - Cure: Antibiotics



Your eye care

Go see a doctor if you think there is something wrong with your eye-

Early detection is essential to keeping damage low and preventing permanent loss of your vision.